Kaan Aykurt

PhD Candidate · AI-driven Networked Systems · Technical University of Munich

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Summary

PhD candidate specializing in **AI for Networking** and **Networking for AI**. I build agentic frameworks and benchmarks for autonomous network operations with Large Language Models (LLMs), and investigate the networking requirements of large-scale AI/ML workloads.

Experience

Chair of Communication Networks, Technical University of Munich — Munich, Germany
Doctoral Researcher — 2022 – present

- Measuring how close TCP gets to RoCE for LLM serving and fine-tuning on Ethernet GPU clusters, showing that it stays within 4% of RoCE in pipeline-parallel serving.
- Distilling small language models to assess their improvement potential for autonomous network management, nearly tripling their success rate on operational networking tasks.
- Built RouteMind, an LLM agent framework that generates BGP configurations and negotiates routing policy between autonomous systems in emulated networks.
- Designed NetLLMBench, an open benchmark that checks LLM-generated network configurations in emulation.
- Analyzed on a hardware testbed how TCP and QUIC behave when data center networks are reconfigured at runtime, and predicted the resulting slowdowns with an ML model that explains 88% of their variance.
- Built and operated a Kubernetes experimental cluster and developed HyPA, a pod autoscaler that adapts to changing request patterns.
- Redesigned the Programmable Communication Networks Lab around containerlab, infrastructure-as-code automation and a class-wide mini-internet project; supervised seven student theses and research projects.

Working Student — 2020 – 2022

- Characterized the network traffic of distributed ML training (PyTorch, MPI) and compared RoCE with TCP.
- Applied genetic algorithms to adversarial benchmarking of data center networks and Kubernetes clusters, generating worst-case workloads that maximize flow completion times and degrade service performance.

IBM — Istanbul, Türkiye
IT Specialist — 2018 – 2019

- Worked on the server monitoring stack of a top-three Turkish bank with IBM Tivoli Netcool and Grafana.

Technical Skills

AI & ML: Python, PyTorch, Hugging Face, vLLM, Distributed Training, LoRA Fine-Tuning, LLM Agents, MCP
Networking: TCP/IP, BGP, OSPF, RDMA, RoCE, InfiniBand, NCCL, P4, FRR, Containerlab
Infrastructure: Linux, Docker, Kubernetes, Helm, Git, GitLab CI, Prometheus, Grafana
Languages: Turkish (native), English (fluent), German (intermediate)

Education

Technical University of Munich, Germany — Expected 2027
Ph.D. in Computer Engineering (Advisor: Prof. Wolfgang Kellerer)

Technical University of Munich, Germany — 2022
M.Sc. in Communications Engineering

Koç University, Istanbul, Türkiye — 2019
B.Sc. in Electrical & Electronics Engineering and B.A. in Economics (Double Major)

Selected Publications & Awards

[Full List on kaanaykurt.me]

- **RouteMind: An LLM-Based Framework for Inter-Domain Routing** — *IEEE GLOBECOM, 2026*
- **NetLLMBench: A Benchmark Framework for LLMs in Networking** — *IEEE NFV-SDN, 2024*
- **On the Performance of TCP in Reconfigurable Data Center Networks** — *IEEE CNSM, 2022*
- **Network Traffic Characteristics of ML Frameworks** — *IEEE CNSM, 2021*
- **3rd Place, Graph Neural Networking Challenge** — *ITU AI/ML in 5G Challenge, 2023*
- **Best Student Paper Award** — *IEEE CNSM, 2022*